IMEN371 Quality Management/Term Project

Integrated Case Study about Korea Cryptocurrency Exchange (CEX) : Coinone

1) QFD of CEX Platforms 2) SPC about BTC/KRW pair's gap

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While the crypto market is rapidly growing, investor protection remains insufficient due to the industry's operational immaturity.

Rising Interest and Competition Among the Top 5 Domestic Exchanges

The crypto market has experienced rapid growth over the past few years, establishing itself as an emerging financial market with a market capitalization reaching $3 trillion as of November 2025.

In South Korea, five virtual asset exchanges have been authorized by the Financial Services Commission (FSC) to offer KRW trading pairs, leading to intense and continuous competition within the domestic market.

Examples of Operational Immaturity in CEXs (Centralized Exchanges)

In November 2025, Upbit suffered a hack worth approximately 50 billion KRW, forcing a prolonged suspension of deposits and withdrawals for certain blockchain assets. This incident resulted in abnormal price formations (price distortions).

KBS 뉴스

[단독] 금융당국, 빗썸 현장조사…"오더북 공유 위법성 검토"

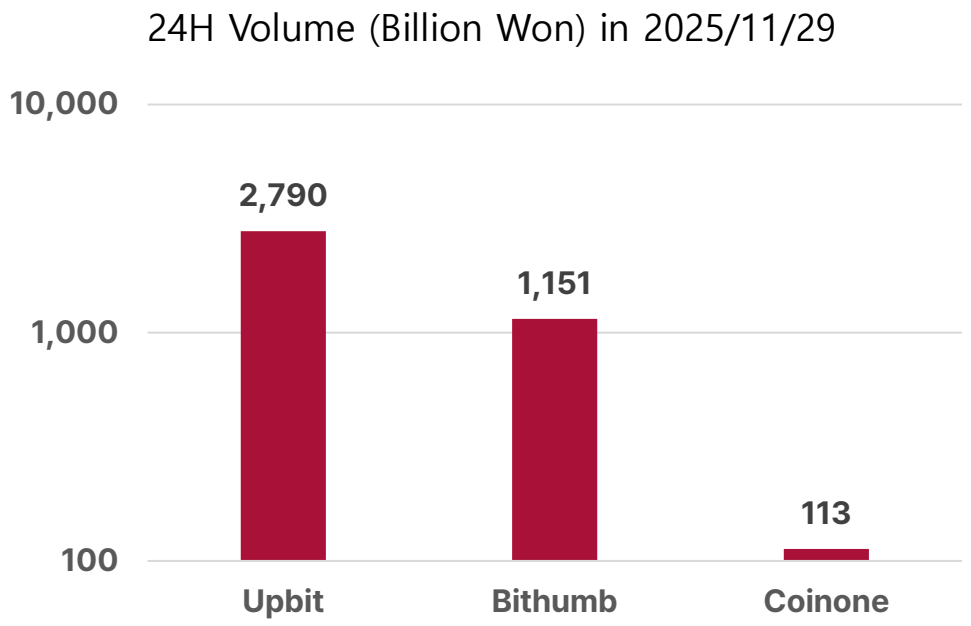
빗썸은 지난달 22일 테더(USDT) 마켓을 열면서, 호주 가상자산거래소 '스텔라'와 오더북을 공유한다고 공지했습니다. 오더북은 가상자산거래소 이용객의...

2025. 10. 1.

In September, Bithumb disclosed that it shared order books for some trading pairs (USDT) with an Australian exchange. Due to potential legal violations, authorities conducted an on-site investigation and search and seizure. Consequently, the affected trading pairs were delisted.

We selected Coinone, an alumni-founded company, as the subject of our analysis. Using a QFD study, we aim to identify critical importance factors and propose strategic improvements.

Korea's Top 3 CEXs: Upbit, Bithumb, and Coinone



Upbit dominates the market with a 24-hour trading volume of approximately 2.7 trillion KRW, followed by Bithumb and Coinone. Coinone currently holds a weaker market position, with significantly lower trading volume compared to the top two competitors.

Selection of Coinone as the Target for Analysis

coinone

가상자산 거래소의 기준을 제시하다

Coinone, led by a university alumnus (CEO Myunghun Cha, CSE 07), currently faces a challenging competitive landscape with a relatively smaller market share.

Based on our case study, we will examine the significant impact of investor protection on the crypto industry and provide actionable recommendations for improvement.

We derived CA through a customer survey, set the degree of importance through AHP, and performed 5-point scale competitive analysis.

Selected CA by Customer Survey / Competitive Comparison for CAs

CA		Importance	Upbit	Bithumb	Coinone	Planned Level	Sales point
Market Quality	Trading liquidity and price stability	5.0	5	4	2	5	1.5
	Exchange scale	3.0	5	4	1	5	1.5
Product Diversity	Diversity of listed coins and markets	1.9	5	5	4	5	1.3
	Additional services	1.3	5	3	3	5	1.3
Cost Efficiency	Trading/withdrawal fees	1.7	5	4	3	5	1.2
Trust & Safety	Platform security level	2.5	5	3	4	5	1.1
	Regulatory compliance and transparency	2.0	5	4	2	5	1.2
User Convenience	UI/UX usability	1.2	5	4	3	5	1.3
	Quality of customer support services	1.1	3	4	3	4	1.2
	Convenience of KRW deposits/withdrawals	1.1	5	4	4	5	1.1

* The competitive analysis evaluation criteria for CA are described in Appendix 2.

AHP Analysis for Importance Weights

Survey Design
- Pairwise comparison scale (1–9) used
- Total 45 pairwise comparisons
- 5 respondents participated

Consistency Check
- Consistency Ratio (CR) = **0.06** < 0.1
  → acceptable consistency

Weight Transformation
- Applied Min–Max normalization (1–5 points)

Input Value Rationale

- The **Planned Level** values were derived by referencing the highest-performing competitor for each item to set a strategic benchmark.
- The **Sales Point** values reflect areas where Coinone is likely to achieve differentiation, based on insights from internal reports and press releases.

Rahman, C. M. L. (2020). *Service Development Strategy Using QFD for Customer Satisfaction in a Bank.* In *Proceedings of the International Conference on Industrial & Mechanical Engineering and Operations Management*, Dhaka, Bangladesh.

We defined the key ECs and evaluated target & conducted a 5-point scale competitive analysis of the ECs based on industry ranges.

EC		Direction	Range	Upbit	Bithumb	Coinone	Target	Competitive Analysis: Upbit	Competitive Analysis: Bithumb	Competitive Analysis: Coinone
Trading Performance & Market Quality	1) # of Supported Trading Pairs	▲	381-659	659	453	381	659	5	4	3
	2) # of System Outages (5yr)	▼	2-42	15	42	2	2	2	1	5
	3) Maker/Taker Trading Fee (%)	▼	0.04-0.25	0.05	0.25	0.20	0.04	4	1	2
	4) Withdrawal Fee (USDT ERC20)	▼	1-4	4	4	3	1	4	4	5
	5) 1y Trading Volume (USD)	▲	0.01B-4.5B	4.5B	1.69B	0.12B	4.5B	5	3	1
User Experience	6) UI/UX Usability Score (points)	▲	3-4	4	3	4	4	4	3	4
	7) Portfolio/Performance Visualization Score (points)	▲	3-4	4	3	3	4	4	3	3
Service Diversity	8) # of Promotional Trading Events (recent 1 month)	▲	6-14	13	14	6	14	4	5	2
	9) # of Value-added Services (staking, lending, etc.)	▲	5-10	6	5	8	10	3	3	5
Investor Protection	10) Customer Support Availability (points)	▲	0-5	2	5	5	5	2	5	5
	11) Education & Investor Training Programs (points)	▲	2-5	5	3	2	5	5	3	2
	12) # of Regulatory Investigations (5yr)	▼	2-7	4	7	3	2	4	2	5
	13) Regulatory Compliance / Risk Warning Clarity (points)	▲	1-4	4	3	2	4	5	4	3
	14) Market Manipulation Prevention Mechanisms (points)	▲	2-3	3	2	2	3	5	4	4

- Directions for each EC were validated through an interview with a domain expert.
- Ranges were calibrated based on actual market values of 5 KRW pair CEXs to reflect realistic industry conditions.
- For each EC, the target was set to the best-performing value among Korean KRW-based CEX platforms (industry leader benchmark).
- Data about EC were collected from CoinMarketCap's exchange category and from data published on each exchange's official website.
- The competitive analysis was conducted by evaluating each EC on a 5-point scale based on expert interviews and consensus from a cross-functional team.

We find correlation of EC elements with Literature Review and Expert Interview.

Correlation of EC elements

EC A	EC B	Correlation of EC A and EC B	P/N
3	5	Higher Fee → Lower Volume	Strong Negative
8	5	More Events → Higher Volume	Positive
9	5	More Services → Higher Volume	Positive
14	5	Stronger Prevention → Higher Volume	Strong Positive
1	5	More Pairs → Higher Volume	Positive
13	12	Clearer Compliance → Fewer Investigations	Negative
2	5	More Outages → Lower Volume	Strong Negative
1	8	More pairs → More events opportunities	Positive
1	3	More pairs → Higher operational cost → Higher fee	Negative
10	5	Better support → Higher trust → Higher volume	Positive
4	5	Higher Withdrawal Fee → Lower Volume	Strong Negative
6	10	Better UI → Fewer Support Requests	Negative
9	8	More Services → More Event Potential	Positive

Literature Review

- Higher trading fees are associated with a statistically significant reduction in trading volume, as users are discouraged from frequent participation.
  - ✓ Chakravarty, S., & Sarkar, A. (1999). Liquidity in U.S. fixed income markets: A comparison of the bid-ask spread in corporate, government and municipal bond markets. SSRN.
- A study found that just two hours of app downtime caused a ~7% drop in purchasing volume over the following two weeks, indicating a strong link between outages and user activity.
  - ✓ Chen, Y., & Biswas, D. (2021). The ripple effect of app service outages: Evidence from mobile commerce. Gies College of Business.
- Redesigning the support portal and improving UI/UX led to a 70% decrease in customer support requests in the Mozilla user community.
  - ✓ Nielsen, J. (2022). Self-service UX: Reducing support load through design. Nielsen Norman Group.

Expert Interview

- An increase in the number of supported trading pairs (EC 1) creates more opportunities for event planning (EC 8), such as new listings and trading competitions, which helps boost user engagement.
- Expanding the number of trading pairs (EC 1) may increase operational costs, which could in turn lead to higher trading fees (EC 3) as a trade-off that requires careful strategic management.

We determined the Relationship Matrix of CAs and ECs through the cross functional team consensus. We confirmed the importance of these factors through the HoQ analysis.

Relationship Matrix

Customer Requirements (A) \ Engineering Characteristics (J)	# of Supported Trading Pairs	# of System Outages (5yr)	Maker/Taker Trading Fee	Withdrawal Fee (USDT ERC20)	Trading Volume	UI/UX Usability Score	Portfolio/Performance Visualization Score	# of Promotional Trading Events (1 month)	# of Value-added Services (staking, lending, etc.)	Customer Support Availability	Education & Investor Training Programs	# of Regulatory Investigations (5yr)	Regulatory Compliance / Risk Warning Clarity	Market Manipulation Prevention Mechanisms
Trading liquidity and price stability	○		▽		◎									◎
Exchange scale	◎				◎			○	▽	▽	▽			
Diversity of listed coins and markets	◎				○				▽					
Additional services									◎		○			
Trading/withdrawal fees			◎	◎	○									
Platform security level		○										◎	○	○
Regulatory compliance and transparency												○	◎	○
UI/UX usability					▽	◎	○							
Quality of customer support services										◎				
Convenience of KRW deposits/withdrawals			▽										○	

Top 3 Weight in CA, EC

Rank	CA	Weight	Coinone
1	Exchange scale/activity	34.7%	3rd of 3
2	Trading liquidity and price stability	28.9%	3rd of 3
3	Regulatory compliance and transparency	9.3%	3rd of 3

Rank	EC	Weight	Coinone
1	Trading Volume	30.3%	3rd of 3
2	# of Supported Trading Pairs	22.1%	3rd of 3
3	Market Manipulation Prevention Mechanism	15.3%	2nd of 3

Process for Determining CA–EC Relationships

- Team members shared their views on the relationships between CAs and ECs.
- An opinion was adopted if at least 3/4 of members agreed.
- The strength of each identified relationship was categorized into three levels (Strong-Medium-Weak).

In both CA and EC aspects, we observed that overall exchange liquidity is considered a critical factor. We also found that Coinone received relatively lower scores in these areas.

* Liquidity: The ability to convert assets into cash quickly and easily without loss of value. Low liquidity increases the risk of unintended financial loss due to factors like slippage and price disparity.

We conducted a secondary analysis using market price data to validate these findings, focusing particularly on Coinone's price disparity (gap) data in the data collection and utilization process.

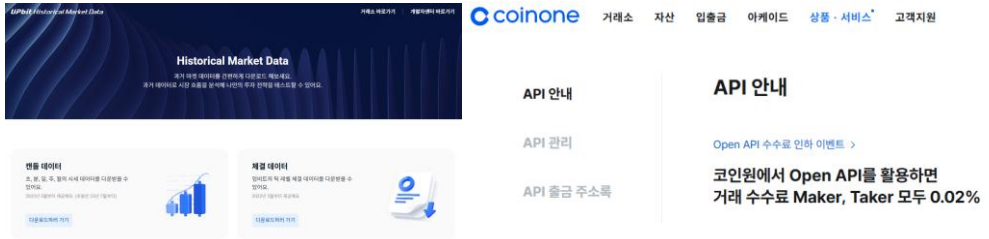
Intuition of Integrated Study

We confirmed that CEX users place significant importance on liquidity. We then sought to verify whether analyzing actual market data from different exchanges would yield conclusions consistent with our QFD study results.

While financial data is typically interpreted using time-series analysis, crypto assets have a **unique characteristic**: the exact **same asset** is often traded at **different prices** across different platforms.

Given this distinction from traditional stocks, we believed we could apply the methodologies learned in class. Consequently, we conducted SPC analysis.

Raw data Gathering & Explanation



Using the CEX API, we collected raw BTC/KRW price data. Each sample consists of 20 data points recorded at 1-hour intervals (09:00 – 04:00), covering the operating hours of both the Korean and U.S. stock markets. **Data measurements were gathered over five weekdays (November 3rd – November 7th) for each sample.**

We calculated the price Gap (%) between Coinone and Upbit (the exchange with the highest liquidity in Korea) to serve as input for the SPC study.

$$\boldsymbol{Gap}\ (\%) = (\boldsymbol{Coinone\ Price} - \boldsymbol{Upbit\ Price})\ /\ \boldsymbol{Upbit\ Price}$$

T-test and ANOVA results for detecting structural bias and time-interval differences in the Gap(%).

one-sample T-test result

Descriptive Statistics

N	Mean	StDev	SE Mean	95% CI for μ
100	-0.03070	0.05382	0.00538	(-0.04138, -0.02002)

μ: population mean of C2

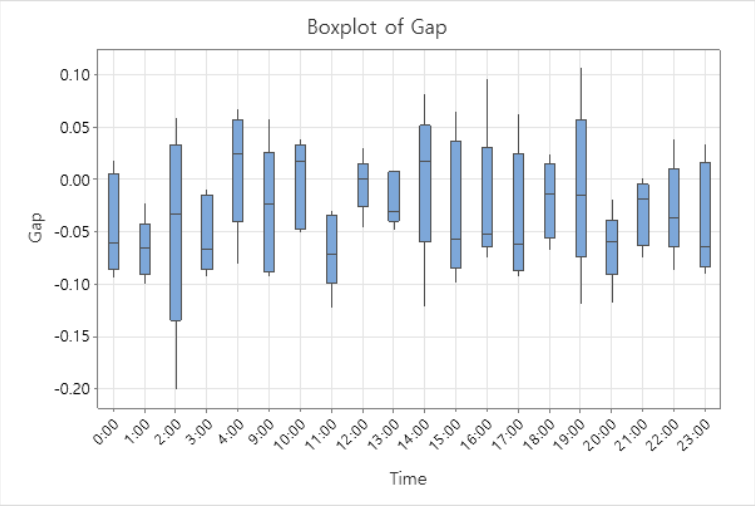
Test

Null hypothesis	H_0: μ = 0
Alternative hypothesis	H_1: μ ≠ 0

T-Value	P-Value
-5.70	0.000

- p-value is less than 0.05, we **reject the null hypothesis** (H_0: mean Gap(%) = 0).
- Because the mean Gap(%) is negative, this indicates a systematic structural bias in which **Coinone is consistently priced lower.**

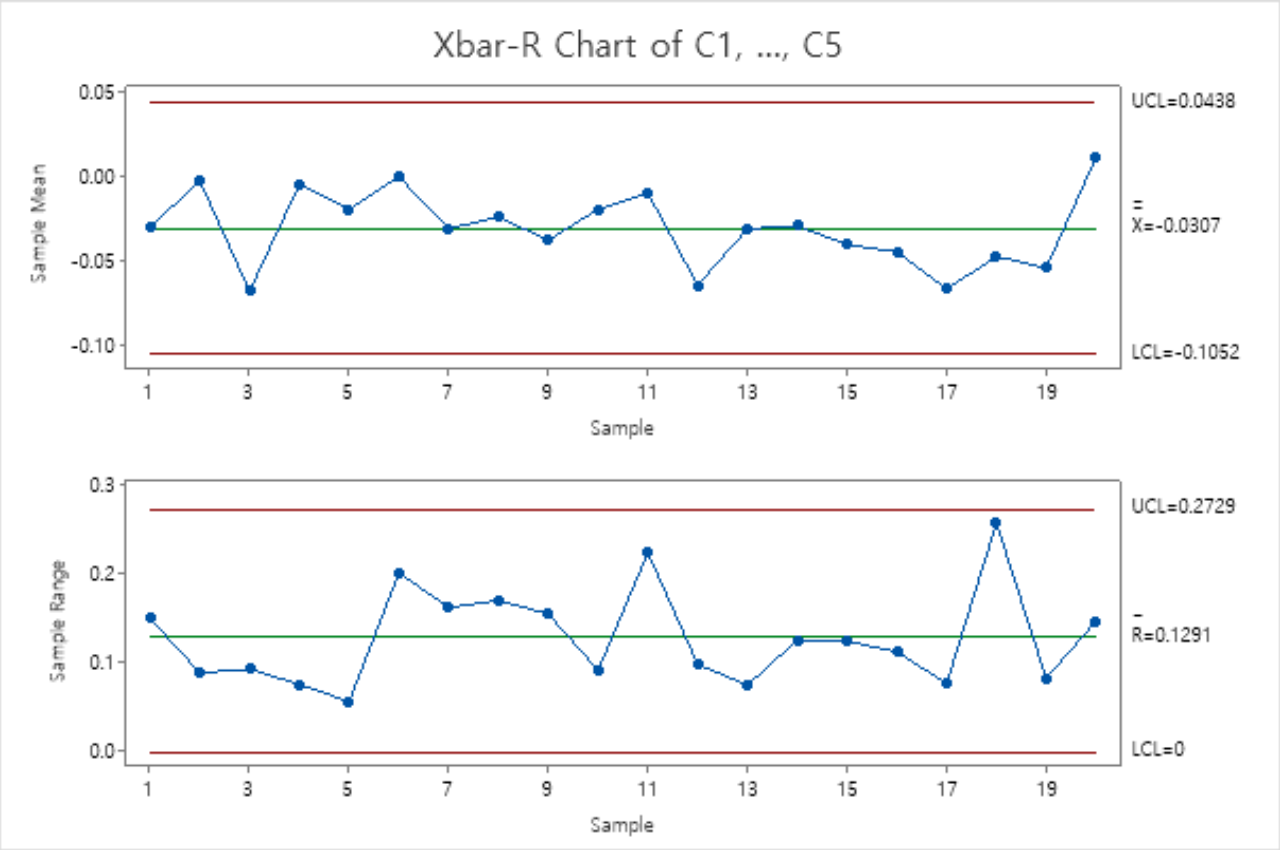
Boxplot & one-way ANOVA result



- **One-way ANOVA across time intervals resulted in F = 0.872 and a p-value of 0.617 (> 0.05).**
- There is **no statistically significant difference** in the mean Gap(%) for each time interval.

Since there was no difference in the mean Gap(%) across time intervals, the microstructure issue is not confined to specific times of day. However, **SPC analysis is still required to assess the stability of the process**.

We conducted $\bar{X}$ – R Control Chart Analysis.

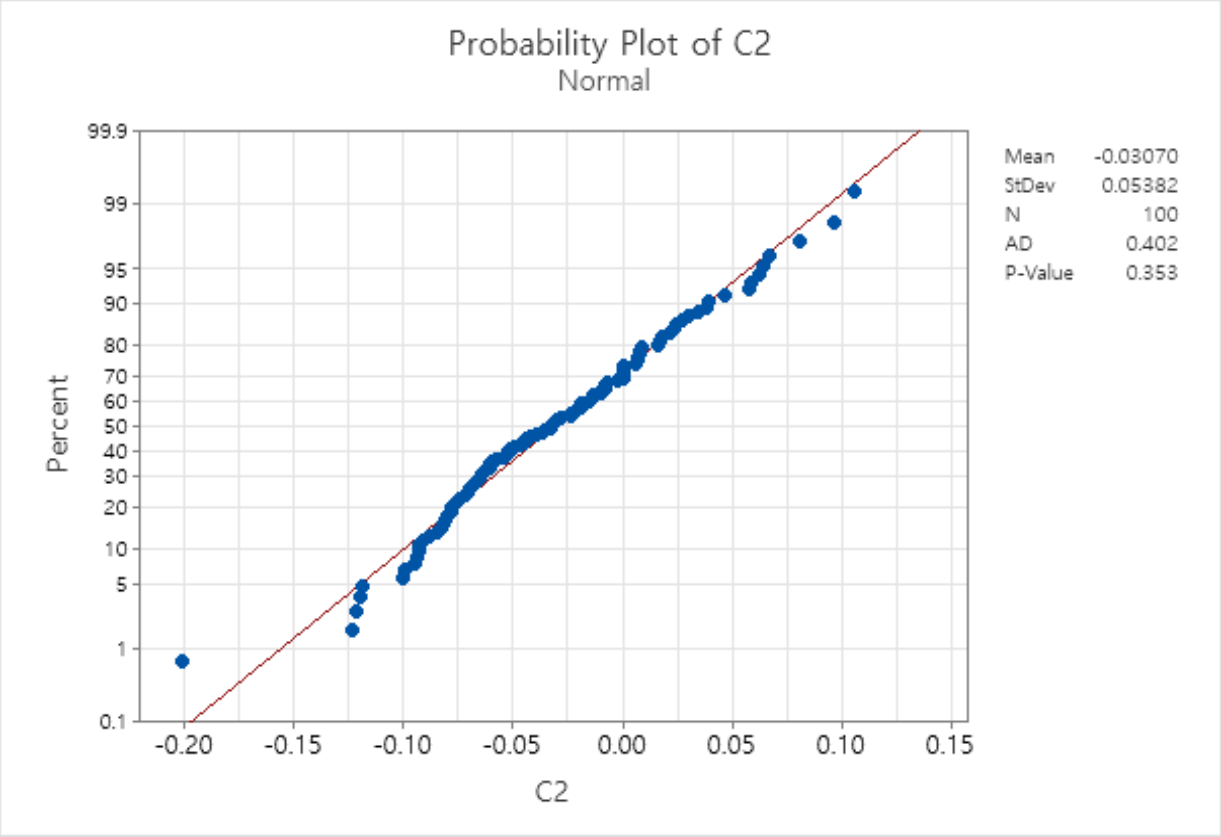


	$\bar{X}$	R
Average	-0.0307	0.1291
UCL	0.0438	0.2729
LCL	-0.1052	0
$\hat{\sigma}$	-	0.0555

- No points exceeded the control limits and no abnormal signals corresponding to Rule 1 and Rule 5 were observed.
- Although there is a slight negative shift in the data, this cannot be interpreted as a special-cause signal and the overall pattern remains stable and consistent.
- Therefore, the process is considered to be in a statistically in-control state.

Applied Rules (for $\bar{X}$ -R Chart):
Rule 1: 1 point > 3 standard deviations from center line
Rule 5: 2 out of 3 consecutive points > 2 standard deviations from center line (same side)

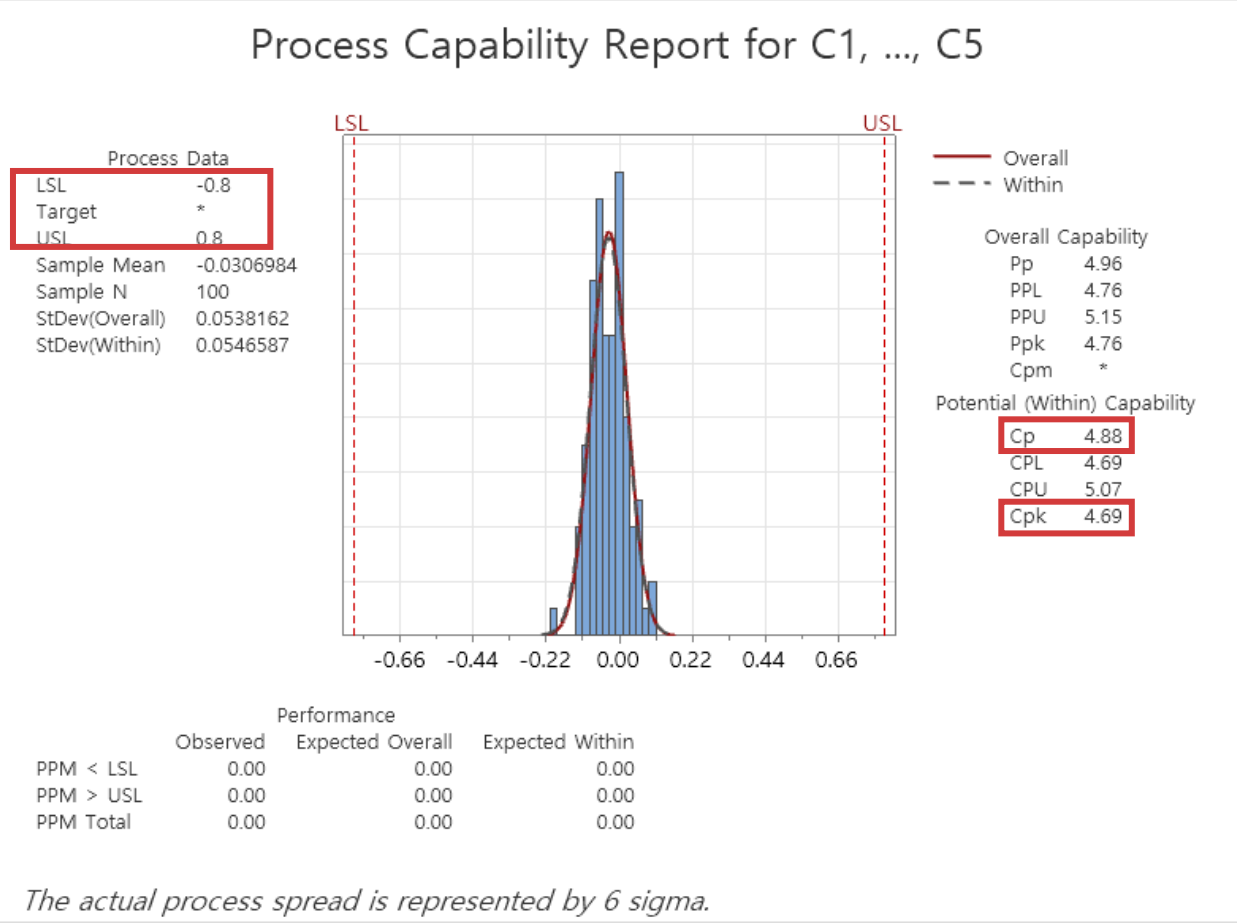
By the result of Anderson-Darling test, the distribution satisfied normality.



- **Anderson–Darling test** has superior power for detecting deviations from normality in the tails, so we use this method.
- H_0: The data follow a normal distribution.
- H_1: The data do not follow a normal distribution.
- The significance level (α) is set to 0.05.
- Since the **p-value** obtained from the normality test is **0.353**, which is greater than 0.05, **we do not reject the null hypothesis (H_0).**

Stephens, M. A. (1974). *EDF statistics for goodness of fit and some comparisons. Journal of the American Statistical Association, 69*(347), 730–737.

We calculated C_p and C_{pk} by Process Capability Analysis.



- The C_p **value of 4.88** indicates extremely low variation within the specification limits, demonstrating excellent process consistency.
- The C_{pk} **value of 4.69**, which far exceeds the Six Sigma benchmark of 1.5, confirms that the process is highly capable and stable.
- The **USL/LSL were set to ±0.8%**, reflecting the threshold at which arbitrage (free lunch) becomes possible when considering BTC withdrawal and deposit fees between the two domestic exchanges.
- The **process mean of –0.03%** lies well within the USL and LSL. Although there is a slight downward shift, the Gap(%) remains sufficiently small and stable to conclude that the process is statistically well-controlled.

Although the BTC/KRW price Gap(%) between the two CEXs shows a slight structural bias, the process is statistically stable.

Data Summary

- There was no significant difference in the mean Gap(%) across time intervals.
- Normality test (p-value = 0.353) confirmed that the adjusted data follow a normal distribution.
- Out of 20 samples, both $\bar{X}$–R charts indicated the **process is in statistical control.**
- Process capability indices: $\boldsymbol{C_p}$ **= 4.88,** $\boldsymbol{C_{pk}}$ **= 4.69**

  → Achieves 6σ quality level.
- All Gap(%) values are between the lower specification limit (LSL=-0.8%) and the upper specification limit (USL=0.8%).

Interpretation & Implication

- Gap(%) is a '**Nominal-the-Best**' quality characteristic with a target value of 0, and its specification limits are set to LSL = –0.8% and USL = 0.8%.
- The process mean of –0.03% is slightly shifted toward the negative side of the target, yet it remains well within the acceptable range.
- The high $C_p$value (4.88) indicates very small process variation, and the fact that C_{pk} (4.69) is slightly lower than C_p reflects a minor downward mean shift; however, the process still demonstrates more than sufficient capability.

The mean Gap(%) shows a downward bias, yet no time-interval differences or SPC abnormal signals appear, indicating a structurally biased but statistically stable process.

Coinone Requires Strategic Marketing Initiatives to Bridge the Gap Between Public Perception and Market Reality.

Suggestions for Coinone

- Need to improved awareness regarding the discrepancy between user expectations and actual market liquidity conditions.
- Example of Marketing Strategy : **Benchmarking Offline Retail**
  - ✓ Offline retailers like **Homeplus** have successfully implemented "Lowest Price Guarantees," where they refund the price difference if a customer finds the same product cheaper at a competitor's store.
  - ✓ **Strategic Application:** If Coinone introduces a compensation program for potential losses caused by liquidity issues, it could attract new investors seeking stability. Since our data shows that the actual price disparity (gap) is minimal, **this strategy would be highly cost-effective while significantly boosting user demand.**

Limitations of Our Study

- Constraints regarding the timeframe of input data.
  - ✓ **Data Granularity Issues:** The study relied on sparse data (hourly representative values) rather than the high-frequency (tick or second-level) data typically available on exchanges.
  - ✓ A more detailed analysis of the **market microstructure** for the BTC/KRW pair gap is required. (It is possible that price disparities could be significantly larger when analyzed at a microscopic level).
- **Need for Expansion to Altcoins**
  - ✓ The study should be extended beyond BTC. While BTC and XRP dominate the domestic market (each accounting for approx. 20% of total volume), the aggregate trading volume of other altcoins is substantial and cannot be ignored.

We identified four core lessons from our integrated QFD and SPC analysis.

Jaena Yoon

I realized that defining EC directions, ranges, and target values is not a simple technical task but a strategic choice that meaningfully influences the analysis. Even small adjustments in these assumptions shifted the HoQ priorities, revealing how sensitive QFD is to the structure set by the analysts. This helped me understand that each modeling parameter must be carefully justified because poorly designed metrics can distort the final strategic conclusion more than the underlying data.

Seoyoung Choi

Through the HoQ process, I learned that QFD does not provide definitive answers but instead generates hypotheses about which technical elements may matter. In our case, the HoQ implied that lower liquidity should lead to pricing instability, yet the SPC analysis contradicted this prediction. This discrepancy revealed that frameworks such as QFD should be treated as tools that structure thinking rather than unquestionable truths, and their outcomes must be critically examined instead of being accepted without verification.

Seungjun Oh

From the SPC analysis, I came to understand the critical distinction between a biased process and an unstable one. Coinone's price structure showed a slight downward bias, yet the system remained statistically in control with no special-cause variations detected across different time intervals. This experience taught me that quality assessment is not determined by whether the mean sits perfectly at the target, but by whether the process behaves in a predictable and stable manner over time.

Hyein Seo

Although Coinone's pricing mechanism demonstrated clear statistical stability, many consumers still associated the platform with elevated risk. This discrepancy made me realize that improving operational performance and improving user perception are fundamentally different challenges. I learned that effective quality management in digital-financial services requires not only technical accuracy but also a deliberate communication strategy that helps bridge the gap between how the system actually performs and how it is perceived.

Appendix 1 - Pairwise comparison matrix used in the AHP analysis

AHP 행렬	거래 유동성 및	거래소 규모 및	상장 코인 및 ㄷ	부가 서비스 및	거래/수수료 구	보안 수준	규제 준수 및 ㅌ	UI/UX	고객지원 서비:	원화 입출금 편	기하평균	중요도	가중치	일관성 벡터	최대고유값	일관성 지수	일관성 비율	min-max
거래 유동성 및 시세	1.00	3.00	5.00	7.00	5.00	3.00	5.00	7.00	7.00	7.00	4.40	0.32	3.38	10.61				5.0
거래소 규모 및 활성.	0.33	1.00	3.00	5.00	3.00	1.00	3.00	5.00	5.00	5.00	2.37	0.17	1.80	10.49				3.0
상장 코인 및 마켓 다	0.20	0.33	1.00	3.00	1.00	0.33	1.00	3.00	5.00	5.00	1.17	0.09	0.91	10.65				1.9
부가 서비스 및 기능	0.14	0.20	0.33	1.00	0.33	0.20	0.33	3.00	3.00	3.00	0.60	0.04	0.49	11.32				1.3
거래/수수료 구조	0.20	0.33	1.00	3.00	1.00	0.33	0.33	3.00	3.00	3.00	0.95	0.07	0.74	10.71				1.7
보안 수준	0.33	1.00	3.00	5.00	3.00	1.00	1.00	3.00	3.00	3.00	1.82	0.13	1.44	10.91				2.5
규제 준수 및 투명성	0.20	0.33	1.00	3.00	3.00	1.00	1.00	3.00	3.00	3.00	1.32	0.10	1.03	10.73				2.0
UI/UX	0.14	0.20	0.33	0.33	0.33	0.33	0.33	1.00	1.00	3.00	0.45	0.03	0.36	10.91				1.2
고객지원 서비스	0.14	0.20	0.20	0.33	0.33	0.33	0.33	1.00	1.00	1.00	0.38	0.03	0.30	10.62				1.1
원화 입출금 편의성	0.14	0.20	0.20	0.33	0.33	0.33	0.33	0.33	1.00	1.00	0.34	0.02	0.27	10.97				1.1
sum	2.84	6.80	15.07	28.00	17.33	7.87	12.67	29.33	32.00	34.00	13.81			107.92	10.79	0.09	**0.06**	

Appendix 2 – Evaluation Basis and Sources of CAs Competitive Analysis

• **Trading Liquidity & Price Stability**

Upbit, ranked among the global top 3 exchanges, records approximately **$1.8 billion** in 24-hour trading volume—by far the highest domestic liquidity—and maintains an average liquidity score of **565**, placing it among the top tier (coinmarketcap.com). With a **70% market share**, its price stability is strong, so it receives **5 points** (chaincatcher.com). Bithumb's trading volume is around **$725 million**, roughly 40% of Upbit's level (coinmarketcap.com), and its liquidity score is lower than Upbit's; thus it receives **4 points**. Coinone's trading volume is only **$103 million**, about 6% of Upbit's (coinmarketcap.com), and its market share is just **3%** (chaincatcher.com), indicating very low liquidity. In 2024, shortly after a new listing, a token on Coinone surged **4,600× within 40 minutes** and then crashed, highlighting extreme volatility driven by poor liquidity (v.daum.net). For these reasons, Coinone is assigned **2 points**.

• **Exchange Size / User Activity**

Upbit has approximately **9.82 million** domestic users and consistently holds a **67–70%** market share, making it the largest platform (chaincatcher.com). Bithumb is smaller, with a **25–30%** market share, placing it in the second tier; therefore, it receives **4 points** (chaincatcher.com). Coinone has about **3.2 million** users and only **3%** market share, resulting in a **1-point** score (chaincatcher.com). (Reference: "The top two exchanges account for 97–98% of the Korean market, while Coinone and others share only 2–3%, showing an extreme oligopoly." – chaincatcher.com)

• **Listed Coins & Market Variety**

All three exchanges list hundreds of coins, offering above-average global diversity. Upbit supports **300+ coins** and **662 trading pairs** across KRW, BTC, and USDT markets (coinmarketcap.com). Bithumb lists **180+ assets** and **280+ trading pairs**, rapidly expanding its altcoin offerings (coinmarketcap.com). Coinone lists **200+ coins**, similar to Bithumb (coinmarketcap.com). However, several major altcoins have been listed late or are still missing compared to Upbit and Bithumb, and the frequency of "investment warning" designations is relatively high. Therefore, it receives **4 points**. (Reference: "As of 2024, Bithumb is surpassing Upbit and Coinone in the number of listed assets, while Upbit and Coinone remain similar in listing count." – chaincatcher.com)

• **Additional Services**

Upbit offers the most diverse services, including an NFT marketplace and staking products. Its NFT platform is the most established among Korean exchanges (coinmarketcap.com), so it receives **5 points**. Bithumb provides limited deposit services, some airdrop events, and basic yield programs but no NFT marketplace or broader ecosystem features; thus **3 points**. Coinone offers PoS-based staking services (coinmarketcap.com) but lacks other innovative features, resulting in **3 points**.

• **Fee Competitiveness**

Upbit charges a flat **0.05%** trading fee—the lowest among major Korean exchanges—earning **5 points** (coinmarketcap.com). Bithumb's base fee is **0.15%**, but with coupons and membership tiers it can be reduced to **0.04%** (coinmarketcap.com). Although the typical user pays slightly more than on Upbit, active users can reduce fees significantly, so it receives **4 points**. Coinone's base fee is **0.2%**, the highest among the three. Maker fees can be reduced to **0%** and taker fees to a minimum of **0.02%**, but these benefits apply mainly to high-volume traders (coinmarketcap.com). For general users, fees remain burdensome; thus it receives **3 points**. Withdrawal fees (e.g., **0.0015 BTC**) are also relatively high, which is another drawback (coinmarketcap.com).

• **Security Level**

Upbit holds ISMS-P, ISO 27001, and other major security certifications, and operates 24/7 AI-based FDS monitoring (coinmarketcap.com). Although it suffered a ~$50M ETH hack in November 2019, it fully compensated users and has had no major incidents since, earning high trust and **5 points**. Bithumb faced multiple security incidents in 2017–2018, including customer information leaks and a ~$30M hack (techcrunch.com). Hot-wallet management issues were also cited. Security has improved since then, but vulnerabilities have been more frequently noted than at Upbit, so it receives **3 points**. Coinone has no notable history of major hacks and implements 2FA, multisig wallets, and continuous monitoring (coinmarketcap.com). Its smaller size may mean fewer attacks rather than inherently superior security, but its clean record is a strength, so it scores **4 points**.

• **Regulatory Compliance & Transparency**

All three exchanges hold ISMS certification and have completed FIU VASP registration. Upbit leads in this category due to strong investor-protection measures, transparent disclosures, and regular quarterly audit reports, earning **5 points**. Bithumb has strengthened internal controls, but past issues involving major shareholders—such as embezzlement and stock-manipulation allegations—create lingering transparency concerns; thus **4 points**. Coinone was involved in a **2023 listing-bribery and market-manipulation case**, in which listing-team executives were indicted for accepting payments to list specific coins and artificially inflating volume and prices. Roughly **25% of listed tokens** were implicated, severely damaging investor trust (cryptoslate.com). Therefore, Coinone receives a very low **2 points**.

• **UI/UX Convenience**

Upbit offers the most polished Korean interface, high mobile-app stability, fast order execution, and superior chart tools—earning **5 points**. Bithumb has improved significantly through UI/UX redesigns, though some menu structures remain inconvenient for users, resulting in **4 points**. Coinone provides Lite/Pro views, but its order-book design and charting tools are simpler, and mobile app ratings are lower than Upbit's—thus **3 points**. (Reference: 38% of users prioritize mobile apps; Coinone's rating is noticeably lower than Upbit's – kraken.com)

• **Customer Support**

Bithumb operates a **24/7 call center and an offline customer center (Gangnam branch)**, earning relatively strong evaluations. With expanded support staffing, it receives **4 points**. Upbit, due to its massive user base, often experiences slower response times. It is improving through online support and an AI chatbot, but lacks in-person service and can be slower in handling issues; thus **3 points**. Coinone offers online and phone support. Although response times can be faster than Upbit due to its smaller scale, its overall support infrastructure is limited; hence **3 points**. (Reference: Prompt and friendly customer support is critical to user trust — ainvest.com)

• **KRW Deposit/Withdrawal Convenience**

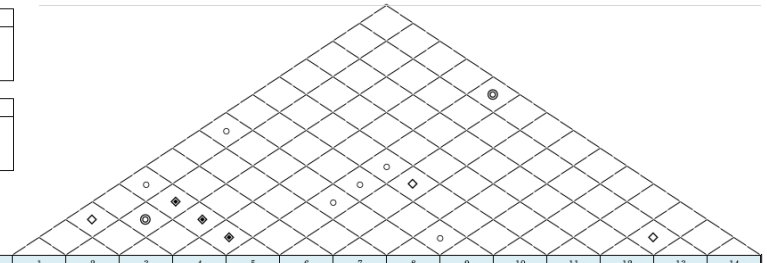
Upbit partners with **K Bank**, enabling 100% remote account issuance and instant transfers—an important competitive advantage since 2018 and still unmatched today (koreatimes.co.kr). For stability and speed, it receives **5 points**. Bithumb and Coinone partner with **NH NongHyup Bank**, which may require in-person branch visits for some users. Once established, transfers are smooth, but onboarding can be more difficult than on Upbit. Bithumb has had isolated withdrawal-delay incidents in the past but has since stabilized. Coinone faced concerns in 2023 regarding contract renewal with NongHyup, raising fears of temporary suspension of KRW transfers, though the issue was resolved. Both Bithumb and Coinone are awarded **4 points**.

Appendix 3 – Final HoQ

Correlations (EC vs EC)	
Strong Positive	◎
Positive	○
Strong Negative	◆
Negative	◇

Relationships (CA vs EC)	
Strong (9)	◎
Medium (3)	○
Weak (1)	▽

Arrows	
Maximize	▲
Minimize	▼
Target	◇

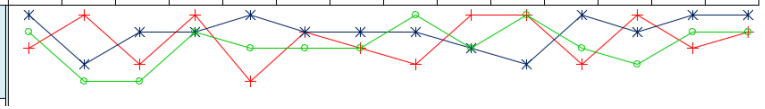


Column #	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Direction of Improvement (P)	▲	▼	▼	▼	▲	▲	▲	▲	▲	▲	▲	▼	▲	▲
Engineering Characteristics (high level)	Trading Performance & Market Quality					User Experience		Service Diversity		Investor Protection				

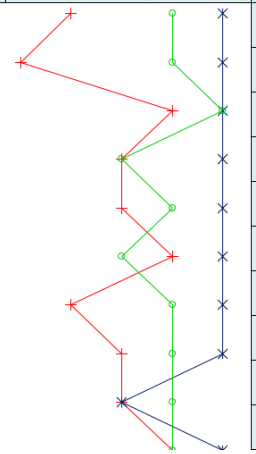
Row #	Customer Requirements (high level)	Customer Requirements (A) / Engineering Characteristics (J)	# of Supported Trading Pairs	# of System Outages (5yr)	Maker/Taker Trading Fee	Withdrawal Fee (USDT ERC20)	Trading Volume	UI/UX Usability Score	Portfolio/Performance Visualization Score	# of Promotional Trading Events (1 month)	# of Value-added Services (staking, lending, etc.)	Customer Support Availability	Education & Investor Training Programs	# of Regulatory Investigations (5yr)	Regulatory Compliance / Risk Warning Clarity	Market Manipulation Prevention Mechanisms	Degree of Importance (B)	Our Product (C)	Competitor #1: UPBIT (C)	Competitor #2: BITHUMB (C)	Planned Level (E)	Important ratio (F)	Sales point (G)	Importance weight (H)	Related Weight (I)
1	Market Quality	Trading liquidity and price stability	○		▽		◎									◎	5	2	5	4	5	2.5	1.5	18.8	28.5%
2	Market Quality	Exchange scale	◎				◎			○	▽	▽	▽				3	1	5	4	5	5	1.5	22.5	34.2%
3	Product Diversity	Diversity of listed coins and markets	◎				○				▽						1.9	4	5	5	5	1.25	1.3	3.1	4.7%
4	Product Diversity	Additional services									◎		○				1.3	3	5	3	5	1.67	1.3	2.8	4.3%
5	Cost Effic	Trading/withdrawal fees			◎	◎	○										1.7	3	5	4	5	1.67	1.2	3.4	5.2%
6	Trust & Safety	Platform security level		○										◎	○	○	2.5	4	5	3	5	1.25	1.1	3.4	5.2%
7	Trust & Safety	Regulatory compliance and transparency												○	◎	○	2	2	5	4	5	2.5	1.2	6.0	9.1%
8	User Convenience	UI/UX usability					▽	◎	○								1.2	3	5	4	5	1.67	1.3	2.6	3.9%
9	User Convenience	Quality of customer support services										◎					1.1	3	3	4	4	1.33	1.2	1.8	2.7%
10	User Convenience	Convenience of KRW deposits/withdrawals			▽										○		1.1	4	5	4	5	1.25	1.1	1.5	2.3%

Importance Weight (M)	4.4	0.2	0.8	0.5	6.0	0.4	0.1	1.0	0.8	0.6	0.5	0.7	1.0	3.0
Related Weight (N)	22.0%	0.8%	3.9%	2.3%	30.1%	1.8%	0.6%	5.2%	3.9%	2.9%	2.4%	3.7%	5.3%	15.1%
Target (R)	659	2	0.04	1	4.5B	4	4	14	10	5	5	2	4	3
Our Product (O)	3	5	2	5	1	4	3	2	5	5	2	5	3	4
Competitor #1: Upbit (O)	5	2	4	4	5	4	4	4	3	2	5	4	5	5
Competitor #2: Bithumb (O)	4	1	1	4	3	3	3	5	3	5	3	2	4	4

Technical Competitive Analysis



Customer Competitive Analysis (D)



Appendix 4 – Data table about Gap(%) (input)

Sample #.	Time	x_1	x_2	x_3	x_4	x_5
		11/3	11/4	11/5	11/6	11/7
1	9:00 AM	-0.00797	0.05777	-0.08475	-0.09284	-0.02371
2	10:00 AM	0.02698	-0.04356	0.0167	0.03825	-0.05132
3	11:00 AM	-0.02954	-0.12296	-0.039	-0.07129	-0.07675
4	12:00 PM	0	-0.04675	0.02957	0	-0.00656
5	1:00 PM	-0.03161	-0.03128	0.00791	-0.04896	0.00656
6	2:00 PM	0.00062	-0.12119	0.01648	0.02145	0.08104
7	3:00 PM	-0.09907	0.06416	0.00856	-0.05769	-0.07019
8	4:00 PM	-0.07464	0.09633	-0.0539	-0.03564	-0.05246
9	5:00 PM	-0.09339	0.06238	-0.01322	-0.0622	-0.08231
10	6:00 PM	0.02375	0.00585	-0.06852	-0.04481	-0.01389
11	7:00 PM	-0.01564	0.10587	-0.02853	-0.11914	0.00729
12	8:00 PM	-0.01926	-0.11819	-0.05996	-0.0596	-0.0644
13	9:00 PM	-0.00995	-0.01881	-0.05226	-0.07484	0.00067
14	10:00 PM	-0.03666	-0.01883	-0.08763	0.03883	-0.0419
15	11:00 PM	0.03422	-0.06503	-0.09077	-0.07776	-0.00265
16	12:00 AM	0.01825	-0.06051	-0.09475	-0.00786	-0.07808
17	1:00 AM	-0.02318	-0.06293	-0.10045	-0.06552	-0.08118
18	2:00 AM	-0.06966	0.00794	-0.03286	-0.20019	0.05816
19	3:00 AM	-0.07992	-0.00988	-0.09307	-0.06706	-0.0209
20	4:00 AM	0.04674	0.06665	0	-0.08105	0.02401

▲ Gap(%)